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Search for New Physics in anomalous jets using jet substructures with the ATLAS detector and LHC Run 3 data

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Sapienza University of Rome

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Introduction

It was the 4 July 2012 when Joseph Incandela, CMS spokesperson, affirmed: "*They combine to give us a combined significance of five standard deviations*". With these exact words, he announced, confirmed by ATLAS spokesperson Fabiola Gianotti a few minutes later, the first observation of the Higgs boson [1, 2], which completed the Standard Model (SM) framework. Fourteen years later, no comparable discovery has followed within the SM paradigm at collider experiments, strengthening the hypothesis that it is the most complete and well-tested description of nature.
In reality, SM probably is not as complete as it appears. In fact, there are many phenomena not included in the framework but confirmed by astrophysical and cosmological observations: among these we can find a quantum description of gravity, Dark Matter [3] and baryon asymmetry. All these processes fall in the category of New Physics (NP) Beyond the Standard Model (BSM). They have been hunted by ATLAS and CMS experiments using traditional data analysis techniques, targeting specific theoretical models but, unfortunately, nothing has been observed yet and only upper limits on production cross-sections were set. There are many reasons for this unexpected outcome: wrong Physical model for experiments, wrong triggering while selecting data, or deviations too small to be observed through normal techniques.
A possible solution to some of these issues could be Anomaly Detection (AD), that is the search for data deviating from what is already known. In our field of interest, Particle Physics, we apply this method to look for anomalies (signal data), deviating from well known reacting and decaying particles (background data). The strength of this method lies in its model agnostic nature, meaning that it does not rely on a specific theoretical model to analyse data. This allows us to explore a more vast range of events, increasing the background rejection efficiency, giving us a first hint of the region where to look more in deep for New Physics with precise data analysis techniques. In this process, unsupervised Machine Learning (ML) helps speed up the process.
This thesis investigates whether such an unsupervised approach can be applied to a concrete search. Using ATLAS official Monte Carlo (MC) samples simulating the ATLAS detector during Run-3, a search for BSM resonant signals decaying into hadronic final states with large-radius jets (used to probe boosted resonance topologies) has been conducted, preparing the needed tools to finalise the analysis on real ATLAS data. Leveraging the Anomaly Score (AS) distributions computed by a Graph Neural Network (GNN) for each jet, the analysis strategy was defined, starting with the definition of Working Points (WPs), thresholds on the AS distribution corresponding to a given background rejection fraction. A crucial part of

the strategy was determining the correct modelling function for the background and signals di-jet invariant mass (m_{JJ}) distribution. After tuning and validating a Double-Sided Crystal Ball (DSCB) function to model the signals and a 5-parameter function for the background distribution, fits to the MC samples were executed, deriving the expected upper limits at 95% Confidence Level on m_{JJ} using the CLs method, at different AS WPs. The study on MC samples was concluded and fully validated by performing sensitivity studies based on a binned maximum likelihood fit to the di-jet invariant mass distribution.
The thesis is structured as follows. In Chapter 1 the Standard Model is presented, together with theories of Physics beyond it. The LHC collider and the ATLAS experiment are discussed in Chapter 2. Chapter 3 presents the Anomaly Detection approach. The typical workflow of a Particle Physics analysis is described in Chapter 4. Chapter 5 presents the analysis conducted during this study and its results. Finally, conclusions are drawn in Chapter 6.

First index draft:

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- Anomaly Detection
  - What is it?
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- How is a Particle Physics Analysis conducted?
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- ADJJ analysis: plots and results

Chapter 1

(Beyond) The Standard Model

The Standard Model (SM) of Particle Physics is a relativistic Quantum Field Theory (QFT) describing three out of four fundamental interactions, strong, weak and electromagnetic (EM), and the behavior of elementray particles under them. For a long time it was attempted to merge the fourth fundamental interaction, gravity, with the SM framework, but those attempts have not been validated yet by experimental observations. Nevertheless, for our purposes, gravity does not play a significant role since the energy scales involved in the processes at a collider experiment, as the LHC, are much lower than the Planck scale, where gravitational effects become relevant.
What is remarkable about the SM is that it is a very well tested theory, with predictions confirmed by experimental observations with a very high precision. The discovery of the Higgs boson in 2012 at the LHC, which completed the SM framework, is a clear example of this.
Despite the success of the SM, there are many phenomena that are not explained by it, but are confirmed by astrophysical and cosmological observations, opening the road for Physics Beyond the Standard Model.
This chapter presentes a brief overview of the SM, highlighting its major features. In addition, hints for Physics Beyond it are exposed.

1.1 An overview of the Standard Model

The fundamental constituents of Universe are precisely described by the SM. They are called elementary particles, which are mathematically described as quantum fields $\psi(x)$ and their dynamics is modelled by Lagrangian densities $\mathcal{L}(\psi, \partial_\mu \psi)$ [4].
Such particles can be graphically pictured as presented in Fig. 1.1. They are divided into two fundamental classes based on their spin quantum number. The first class is composed by *fermions*, which cover the first three column of Fig. 1.1 and are organised in three *generations* of increasing proper mass; they have spin 1/2 and constitute *matter*. The second class is those of *bosons*, with integer spin, which *mediate* fundamental interactions and generate mass.
Fermions are further classified into two distinct categories depending on their participation in the strong interaction. The first category is relative to *quarks*. They carry *colour charge* (red, green, blue), making them triplets under $SU(3)$ and therefore interact via all fundamental forces, among which the strong one. Quarks

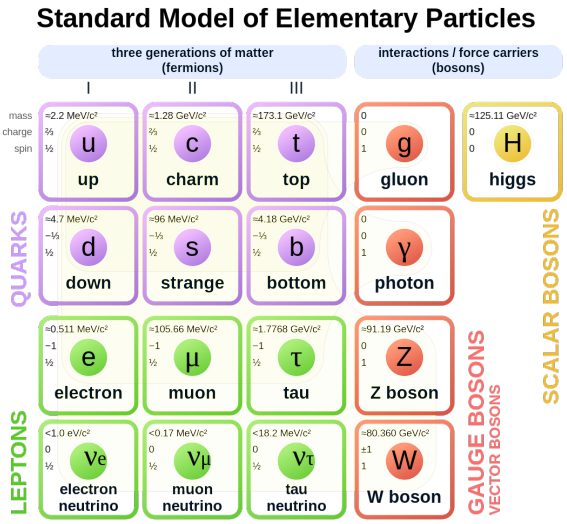


Figure 1.1 The SM framework. Figure from Ref. [5].

exist in six flavors, organized into three generations of weak SU(2) doublets:

$$\begin{pmatrix} u \\ d \end{pmatrix}, \quad \begin{pmatrix} c \\ s \end{pmatrix}, \quad \begin{pmatrix} t \\ b \end{pmatrix} \tag{1.1}$$

Quarks carry a fractional electromagnetic charge: the *up-type* ones (u, c, t) have a charge of $+2/3$, while the *down-type* ones (d, s, b) carry a $-1/3$ charge.
The second category of fermions is associated to *leptons*; they are six and similarly grouped into three generations:

$$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix}, \quad \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix}, \quad \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix} \tag{1.2}$$

The charged leptons (e^-, μ^-, τ^-) carry an electric charge of -1 and interact via electromagnetic and weak forces. Neutrinos $(\nu_e, \nu_\mu, \nu_\tau)$ are electrically neutral and interact exclusively via the weak force.
Both quarks and charged leptons, have their own anti-particles, which have the same mass of the matter components but opposite quantum numbers. Regarding neutrinos, it is still debated and to be proved whether they have a distinct anti-particles or they are Majorana fermions, being their own anti-particles.
The second class of particles is associated to mediatiors and mass. Spin 1 vector bosons, known as *gauge bosons*, are the mediators of three of the fundamental interactions, which arise naturally from the requirement of local gauge invariance. Among those fields we find the 8 massless gluons g mediating the strong interaction, the massless photon γ is associated to electromagnetic interaction and 3 massive vector bosons, $W^\pm$ and Z^0, with masses $m_W = 80.36$ GeV, $m_Z = 91.19$ GeV, mediating the weak interaction.
In conclusion, a spin 0 scalar particle, the *Higgs boson* H, with a mass $m_H = 125.13$ GeV, completes the particle spectrum, arising as a physical state from the

spontaneous breaking of electroweak symmetry.

The SM is a gauge theory, meaning that the Lagrangian is invariant under local transformations of the fields. The Lagrangian density describing the SM dynamics can be written as:

$$\mathcal{L}_{\text{SM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - i\bar{\psi}\gamma_\mu D^\mu\psi + (D_\mu\phi)^\dagger(D^\mu\phi) - V(\phi) + \bar{\psi}_i y_{ij}\psi_j\phi + h.c., \quad (1.3)$$

The gauge group of the SM, under which such lagrangian is invariant, is a non-abelian symmetry group defined as:

$$SU(3)_C \otimes SU(2)_L \otimes U(1)_Y \quad (1.4)$$

where $SU(3)_C$ describes the strong interaction via Quantum Chromodynamics (QCD) with subscript C standing for colour, and $SU(2)_L \otimes U(1)_Y$ represents the unified electroweak (EW) interaction, with L associated to the left component of fermions and Y to hypercharge.
The terms of the equation 1.3 above are presented in detail in the following sections.

1.1.1 Quantum Electrodynamics

The first and simplest gauge theory within the SM framework is Quantum Electrodynamics (QED). It describes the electromagnetic interactions between a charged massive fermion, with mass m and represented by a field $\psi(x)$, and the photon, the massless gauge boson represented by the vector field A_μ. Using the Lagrangian formalism [6], QED is constructed starting from the free Dirac Lagrangian for a massive fermion [7]:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi \quad (1.5)$$

In the equation above, $\bar{\psi} = \psi^\dagger\gamma^0$ is the Dirac adjoint of the field ψ and γ^μ are the Dirac matrices derived from Pauli matrices σ_i (i =1,2,3 and μ=0,1,2,3):

$$\gamma^0 = \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \quad (1.6)$$

The Dirac Lagrangian 1.5 is invariant under a global $U(1)$ phase transformation $\psi(x) \rightarrow e^{ie\alpha}\psi(x)$, with e the electric charge of the fermion and α a constant phase. Such invariance is not preserved under a local $U(1)$ phase transformation $\psi(x) \rightarrow e^{ie\alpha(x)}\psi(x)$, where $\alpha(x)$ is a space-time dependent phase. In order to preserve the invariance, the partial derivative ∂_μ is replaced by the covariant derivative D_μ:

$$D_\mu = \partial_\mu + ieA_\mu(x) \quad (1.7)$$

where $A_\mu(x)$ is a vector gauge field representing the photon field, which transforms as $A_\mu(x) \rightarrow A_\mu(x) - \partial_\mu\alpha(x)$. Including the gauge-invariant kinetic term to take into account for the new field A_μ, we derive the full QED Lagrangian:

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad (1.8)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor. The interaction term $-e\bar{\psi}\gamma^\mu\psi A_\mu$ rules the coupling of charged fermions to photons with a strength proportional to the electric charge e.
The invariance under a local $U(1)$ transformation gurantees, via the Noether's theorem [8], the existence of a conserved current and charge which, in the case of QED, are the electromagnetic current and the electric charge, respectively.

1.1.2 Quantum Chromodynamics

The building of the Lagrangian for QCD is similar to the one for QED: the Dirac Lagrangian is required to be invariant under a local symmetry group, which in this case is the non-Abelian $SU(3)_C$ group, where C represents the triplet color degree of freedom (red, green, blue), the conserved charge of QCD [9]. This theory models the strong interaction between *quarks*, fundamental constituents of hadrons represented as fermionic fields $q(x)$, and *gluons*, the massless gauge bosons represented as vector fields $G^a_{\mu\nu}(x)$ with $a = 1, ..., 8$. Similarly to QED, the local transformation of the quark field is defined as $q(x) \to q'(x) = e^{ig_s T^a \alpha_a(x)} q(x)$, with g_s the strong coupling constant and $\alpha_a(x)$ the local phase. The term T^a appearing in the previous transformation equation is associated to the generators of the $SU(3)_C$ group, which can be expressed via the Gell-Mann matrices as $T^a = \frac{1}{2}\lambda^a$; such generators satisfy $[T^a, T^a] = if^{abc}T^c$ with f^{abc} being the structure constants of $SU(3)$.
Also in this case, a covariant derivative is introduced to preserve the invariance of the Lagrangian under local transformations:

$$D_\mu = \partial_\mu + ig_s T_a G^a_\mu \tag{1.9}$$

The non-Abelian structure of the group, gives rise to the QCD field tensor:

$$F^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu - g_s f^{abc} G^b_\mu G^c_\nu \tag{1.10}$$

The presence of the non-linear structure factor f^{abc} both in the commutators and in the tensor, is associated to a particular feature of QCD: the triple and quadruple gluon self-interaction of gluons, a feature not present in QED.
Finally, the Lagrangian describing the dynamics of QCD is written as:

$$\mathcal{L}_{\text{QCD}} = \bar{q}\left(i\gamma^\mu D_\mu - m\right) q - \frac{1}{4} F^a_{\mu\nu} F^{a,\mu\nu} \tag{1.11}$$

The term g_s, cited before in the local transformation, is usually presented as $\alpha_s = \frac{g_s^2}{4\pi}$, which is the running coupling constant of QCD, whose value depends on the energy scale Q^2 of the process. The non-Abelian nature of QCD gives rise to a peculiar behavior of $\alpha_s(Q^2)$. In fact, at high energy scales (meaning short distances), $\alpha_s(Q^2)$ decreases logrithmically, giving rise to the *asymptotic freedom*, enabling accurate calculations via perturbative QCD (pQCD). On the other hand, at low energy scales (meaning long distances), $\alpha_s(Q^2)$ increases, giving rise to the *color confinement*, which prevents colored particles from existing as isolated free states. In this regime, quarks and gluons undergo fragmentation and hadronization, producing collimated sprays of energetic hadrons, known as *jets*.

1.1.3 Electroweak model

1.2 Hints of Physics Beyond the Standard Model

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1.2.1 Electroweak Model

The unification of electromagnetic and weak interactions is realized through the Glashow-Weinberg-Salam (GWS) model, structured around the local gauge symmetry group $SU(2)_L \times U(1)_Y$. The left-handed fermion components transform as doublets under weak isospin T, whereas right-handed components transform as singlets:

$$\Psi_L = \begin{pmatrix} \nu_L \\ e_L^- \end{pmatrix} \quad (T = 1/2, T_3 = \pm 1/2), \qquad e_R^- \quad (T = 0, T_3 = 0) \tag{1.12}$$

Electric charge Q, weak isospin T_3, and weak hypercharge Y are unified by the Gell-Mann-Nishijima relation:

$$Q = T_3 + \frac{Y}{2} \tag{1.13}$$

The gauge-covariant derivative introducing three W_μ^i $(i = 1, 2, 3)$ bosons for $SU(2)_L$ and one B_μ boson for $U(1)_Y$ is written as:

$$D_\mu = \partial_\mu - igT^i W_\mu^i - ig'\frac{Y}{2}B_\mu \tag{1.14}$$

Spontaneous Symmetry Breaking (The Higgs Mechanism) Unbroken $SU(2)_L \times U(1)_Y$ symmetry forbids mass terms for gauge bosons and fermions, as explicit mass terms violate local gauge invariance. Mass generation is accommodated via the Brout-Englert-Higgs mechanism by introducing a complex scalar doublet field Φ:

$$\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \quad \mathcal{V}(\Phi) = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2 \quad (\mu^2 < 0, \lambda > 0) \tag{1.15}$$

The scalar field acquires a non-zero vacuum expectation value (VEV) $v = \sqrt{-\mu^2/\lambda} \approx 246$ GeV, spontaneously breaking the symmetry:

$$SU(2)_L \times U(1)_Y \xrightarrow{\text{SSB}} U(1)_{\text{EM}} \tag{1.16}$$

Expanding Φ around its minimum in the unitary gauge transforms three degrees of freedom into longitudinal polarization states for the gauge physical bosons $W^\pm$ and Z^0:

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2), \quad \begin{pmatrix} Z_\mu \\ A_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & -\sin\theta_W \\ \sin\theta_W & \cos\theta_W \end{pmatrix}\begin{pmatrix} W_\mu^3 \\ B_\mu \end{pmatrix} \tag{1.17}$$

where $\theta_W = \arctan(g'/g)$ is the weak mixing angle. The photon field A_μ remains massless, while gauge bosons acquire masses $m_W = \frac{1}{2}gv$ and $m_Z = \frac{1}{2}v\sqrt{g^2 + g'^2}$. Fermions acquire mass through gauge-invariant Yukawa couplings:

$$\mathcal{L}_{\text{Yukawa}} = -y_f\bar{\Psi}_L\Phi f_R + \text{h.c.} \quad \Longrightarrow \quad m_f = \frac{y_f v}{\sqrt{2}} \tag{1.18}$$

1.3 Hints of Physics Beyond the Standard Model

Despite the extraordinary predictive precision of the Standard Model—culminating in the discovery of the Higgs boson at the LHC in 2012 [**?** **?**]—it cannot be regarded as a complete theory of fundamental interactions. Unresolved theoretical open issues and physical phenomena observed in nature mandate the existence of Physics Beyond the Standard Model (BSM) [**?**]:

- **Dark Matter:** Astrophysical and cosmological observations (galaxy rotation curves, gravitational lensing, cosmic microwave background) firmly establish that non-baryonic *Dark Matter* comprises $\approx 26.8\%$ of the energy density of the Universe [**?**]. The SM offers no suitable stable, cold, weakly-interacting candidate particle [**?**].
- **Neutrino Masses:** Oscillations observed in solar, atmospheric, and reactor neutrinos demonstrate that neutrinos possess non-zero masses, contrasting the strict masslessness assumed in the minimal SM structure.
- **Baryon Asymmetry:** The observable universe consists almost entirely of matter rather than antimatter [**?**]. The magnitude of CP violation provided by the SM CKM matrix is insufficient to account for the observed cosmic matter-antimatter asymmetry [**?**].
- **The Hierarchy Problem:** Quantum loop corrections to the scalar Higgs boson mass quadratic divergences yield corrections proportional to the Planck scale ($\Lambda_{\text{Planck}} \approx 10^{19}$ GeV). Maintaining $m_H \approx 125$ GeV requires extreme, fine-tuned cancellations without a protective symmetry (such as Supersymmetry).
- **Quantum Gravity:** The Standard Model does not incorporate gravitational interactions. A quantum formulation of gravity compatible with General Relativity remains omitted at high energy scales.

These limitations motivate the search for new BSM resonance states and heavy new physics at collider experiments like the Large Hadron Collider (LHC)[cite: 9].

Chapter 2

The LHC and the ATLAS experiments

2.1 The LHC

Chapter 3

Anomaly Detection

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Chapter 4

A particle physics analysis

4.1 The LHC

Chapter 5

ADJJ analysis

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